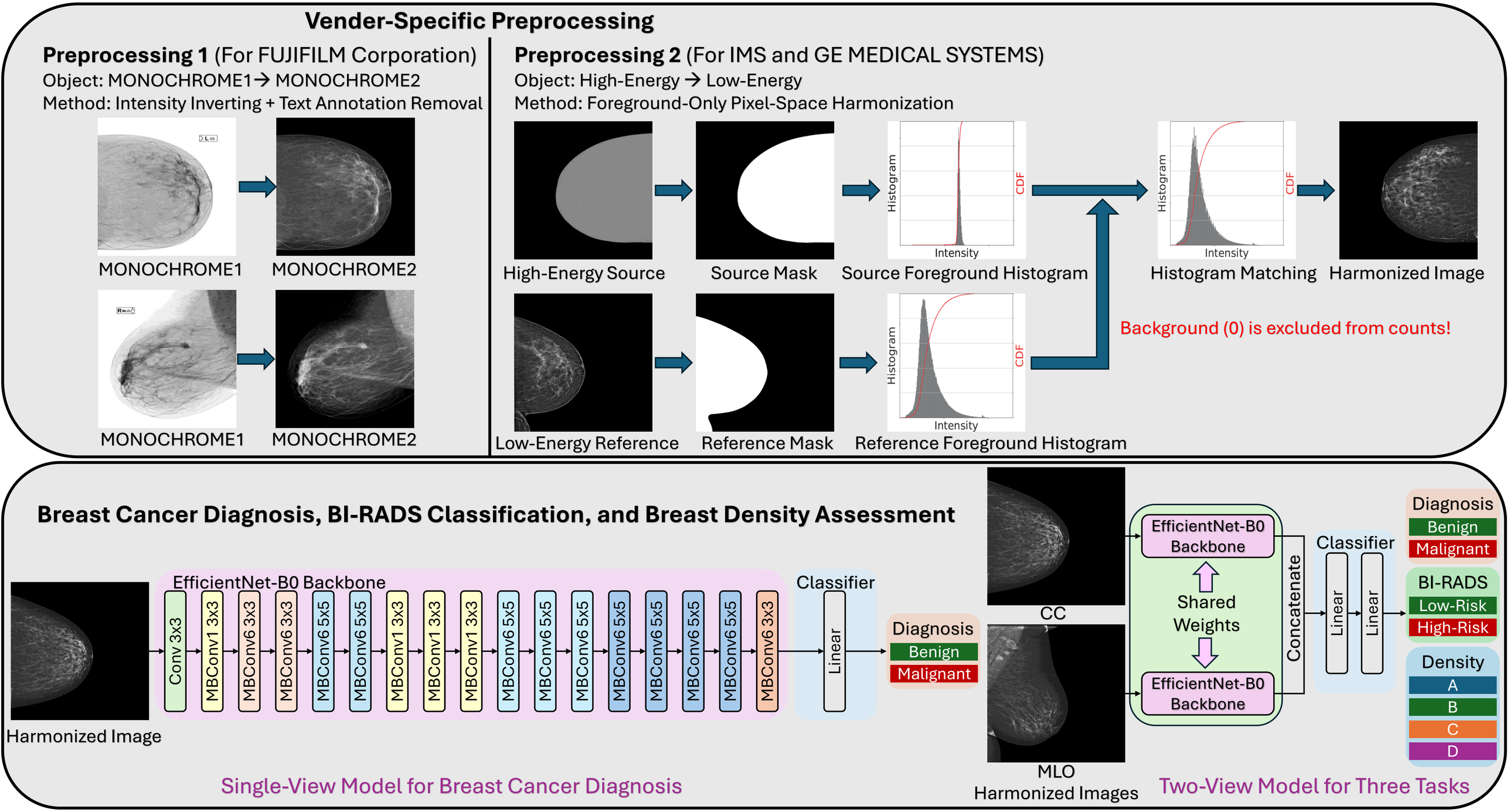
Vender-Specific Preprocessing
Preprocessing 1 (For FUJIFILM Corporation)
Object: MONOCHROME1→ MONOCHROME2
Method: Intensity Inverting + Text Annotation Removal
MONOCHROME1
MONOCHROME2
MONOCHROME1
MONOCHROME2
Preprocessing 2 (For IMS and GE MEDICAL SYSTEMS)
Object: High-Energy → Low-Energy
Method: Foreground-Only Pixel-Space Harmonization
High-Energy Source
Source Mask
Histogram
CDF
Intensity
Source Foreground Histogram
Histogram Matching
Harmonized Image
Low-Energy Reference
Reference Mask
Reference Foreground Histogram
Background (0) is excluded from counts!
Breast Cancer Diagnosis, BI-RADS Classification, and Breast Density Assessment
Harmonized Image
EfficientNet-B0 Backbone
Conv 3x3
MBConv1 3x3
MBConv6 3x3
MBConv6 5x5
Classifier
Linear
Diagnosis
Benign
Malignant
Single-View Model for Breast Cancer Diagnosis
CC
MLO
Harmonized Images
EfficientNet-B0 Backbone
Shared Weights
Concatenate
Classifier
Linear
Diagnosis
Benign
Malignant
BI-RADS
Low-Risk
High-Risk
Density
A
B
C
D
Two-View Model for Three Tasks